Richmond Baidoo (MSc Land Management and Geospatial Science | TUM)
Rexford Boafo Asiedu (MSc Land Management and Geospatial Science | TUM)
Mohammadisharmeh Samaneh (MSc Land Management and Geospatial Science | TUM)
Yuanqi Cheng (MSc Information Technology for Built Environment | TUM)
Ritthaisong Korait (MSc Computer Science | HM)
Gulsah Deniz (MSc Environmental Science | TUM)

sustAInability Project Report

CivicTwin Munchen

Your AI-powered Digital Self in the City

This project holistically addresses how a city administration can use AI to improve public application processing while safeguarding fairness, legitimacy, accountability, citizen trust, and ecological sustainability. In the DAICE challenge, AI-assisted citizen services are evaluated not only by efficiency gains but also by their legal and ethical limits, their impact on caseworkers, their transparency to citizens, and their ecological resource use. A central pain point identified is that Munich has many AI ideas across different departments, but no structured method to prioritize which ideas should be implemented first, limited, postponed, or rejected. The city also lacks a long-term orientation for deciding where AI should and should not be used in public administration. This study draws on the academic knowledge and expertise of members from diverse fields, including environmental engineering, computer science, and land and geospatial sciences. As such, our solution is not "how to automate bureaucracy as much as possible," but rather how to create a practical decision matrix and framework that integrates fragmented AI services, and helps the city evaluate AI use cases based on legal risk, public value, social fairness, administrative responsibility, and ecological cost.

1.0 Introduction

Munich is a leader in digital transformation, ranking first in the 2024 Bitkom Smart City Index and offering 134 digital services, including AI-supported tools, across 15 departments. Despite this progress, significant segments of the population, such as elderly residents, newcomers, and students, face persistent obstacles related to language, system complexity, and fragmented service delivery. The city's current AI initiatives, while promising, remain siloed and insufficiently personalized, with limited multilingual support and inadequate responsiveness to diverse user needs.

Navigating municipal services remains complex, particularly for those unfamiliar with administrative procedures. Even basic requests can require managing multiple interconnected tasks across departments, forms, and languages. Although Munich has invested in AI systems and a comprehensive service portal, these tools are largely task-specific and do not provide an integrated, user-centered experience. As a result, the public administration process is perceived as fragmented, both by users and within the administration itself.

The central challenge is to develop a responsible framework that unifies existing AI systems into a citizen-centered service journey. Isolated improvements are insufficient if the overall experience remains disjointed. This effort must comply with the EU AI Act and GDPR, ensuring risk management, transparency, and human oversight (European Commission, n.d.; European Parliament & Council of the European Union, 2024). Also, the proposed framework facilitates equitable access to services while upholding principles of legality, accountability, sustainability, and social benefit, consistent with international standards for trustworthy AI.

1.1 Key Problem Dimensions

- ✓ *Citizen-side fragmentation:* Citizens must translate life events into separate municipal tasks, offices, documents, and appointments.
- ✓ *City-side fragmentation:* Existing AI systems are useful but disconnected, which limits their value for end-to-end citizen journeys.
- ✓ *Trust and legal sensitivity:* Public-sector AI must remain transparent, contestable, and human-supervised where rights or legal status may be affected.
- ✓ *Inclusion and accessibility:* Language, administrative literacy, disability, and digital skills shape whether citizens can actually use public services.
- ✓ *Sustainability and public value*: The city should reuse existing AI investments and avoid unnecessary model building, energy use, and institutional complexity.

2.0 Problem Analysis

Our study refers to existing literature grouped into four strands: risk-based AI regulation, data protection and automated decision-making, German administrative law, and sustainable AI.

The EU AI Act provides the core European risk-based framework for AI systems, which is important because municipal AI use cases may involve public access to services, citizen vulnerability, or risks to fundamental rights. The GDPR regulates the processing of personal data and is especially relevant when AI systems use citizen profiles, application histories, language preferences, or other personal information.

The Article 29/EDPB guidance on automated individual decision-making and profiling helps interpret GDPR-related risks, including transparency, safeguards, and individual rights, in automated or AI-assisted contexts. In German administrative law, VwVfG §35a permits fully automated administrative acts only under specific legal conditions, while VwVfG §39 requires reasons in administrative decisions.

Sustainable AI literature is relevant because it distinguishes “AI for sustainability” from the “sustainability of AI,” meaning that an AI system should be assessed across its full lifecycle rather than assumed to be sustainable simply because it digitizes a service. The SCAIS framework also operationalizes sustainable AI through a broad set of ecological, social, economic, and governance criteria and indicators.

Munich’s own citizen service and online service reiterate that the city already offers many services digitally and therefore needs an evaluation model applicable to real municipal service processes. With these perspectives on data literacy, analysis, and problem understanding, we propose a decision matrix and a prototype to address the challenge.

3.0 Multi-modal Solution

3.1 Framework Overview & 6-Layer Architecture

3.2 The Multi-Criteria Decision Matrix

The governance framework uses a decision matrix to evaluate AI use cases before implementation. The purpose of this matrix is to ensure that AI systems comply with legal requirements, ethical standards, and governance principles in a structured manner.

To do this, each AI proposal is evaluated across seven dimensions:

- ✓ *Purpose & Public Value*: Does the AI solve a real public service problem?
- ✓ *Fairness:* Does the AI treat all citizens fairly without bias?
- ✓ *Transparency:* Can the AI's decisions be explained?
- ✓ Accountability: Is there a clear person or organization responsible for the system?
- ✓ *Privacy & Data Protection:* Does the AI comply with GDPR and protect personal data?

- ✓ *Human Oversight:* Can humans monitor, review, or override AI decisions?
- ✓ *Sustainability:* Is the AI environmentally, socially, and economically sustainable?

Each dimension is scored from 1 (Poor) to 5 (Excellent). The overall score determines the risk level and the required action for the city administration, as shown in the matrix below:

Overall Score	**Risk Level**	**Recommendation**	**Action**
30–35	🟢 Low Risk	Ready for Deployment	AI meets governance requirements and can be deployed.
20–29	🟡 Medium Risk	Deploy with Improvements	AI can be deployed after addressing identified weaknesses and implementing additional safeguards.
Below 20	🔴 High Risk	Not Recommended	AI should not be deployed until major governance, ethical, legal, or technical issues are resolved.

3.3 Comparative Use Case Evaluation: CivicTwin vs. Standard Municipal AI

To demonstrate how the proposed matrix works in practice, the CivicTwin prototype is compared against two standard AI tools that the City of Munich currently considers or uses: Zammad AI and MUCGPT.

3.3.1 Zammad AI (Ticket Routing)

Zammad AI is used to optimize the internal routing of citizen service tickets. Under our framework, it scores in the Low-Risk category (30–35 points). Because it is a backend automation tool that only handles metadata, it does not evaluate citizens or make legal decisions. It has very few data privacy risks and requires low energy consumption. It meets the governance requirements and is ready for deployment.

3.3.2 MUCGPT (Text Support)

MUCGPT is an internal tool that helps administrative employees draft texts and summarize documents. Our matrix classifies this as a Medium Risk system (20–29 points). While they provide high public value by saving caseworkers time, large language models carry risks of hallucinations and data leaks. Therefore, it requires additional safeguards before deployment. Specifically, a human caseworker must review and approve all AI-generated text to ensure accountability.

3.3.3 CivicTwin (Personalized Citizen Assistant)

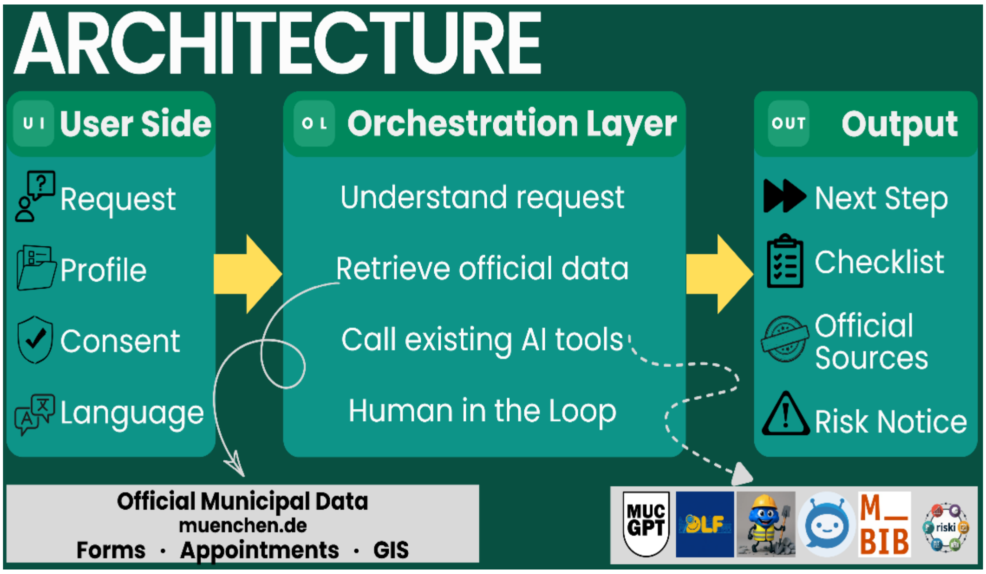
CivicTwin is a highly complex yet scalable, citizen-facing assistant that handles sensitive personal profiles and life events. Because of the high data privacy and legal risks, a standard deployment of this tool would score below 20 (High Risk) and be rejected.

However, by applying our 6-layer architecture, CivicTwin adds the necessary safeguards to reach the Medium Risk (20–29 points) category. It achieves this through three main protections:

- ✓ *Privacy:* The user completely owns their profile data locally, and the city holds no permanent copy, satisfying GDPR.
- ✓ *Oversight:* The AI only retrieves information and provides guidance. It cannot make automated legal decisions, which complies with German administrative law (VwVfG §35a).
- ✓ *Sustainability:* The energy footprint of the AI is justified by the environmental savings of reducing unnecessary physical trips and paper use.

This comparison shows that even complex AI ideas can be implemented responsibly when guided by a rigorous evaluation framework.

3.4 Implementation of CivicTwin



CivicTwin is the proposed solution, which comes off as a municipal AI orchestration framework for citizen services in Munich. Rather than replacing existing systems or building a new large AI model, it integrates citizens, municipal data, current AI tools, databases, and human oversight into a unified service journey.

Inspired by digital twins in infrastructure management, CivicTwin applies this concept to the citizen journey (Fuller et al., 2020). It does not replicate individuals, but creates a consent-based digital representation of each citizen's administrative context, current goal, language, location, deadlines, documents, service history, and consent status, making bureaucracy structured and navigable.

Citizens interact with CivicTwin by entering questions or setting up a profile. Based on user input, the system produces a personalized administrative roadmap that includes checklists, official links, document requirements, appointment guidance, multilingual support, and access to human assistance. Behind the interface, CivicTwin works as a toolbox framework. First, the system interprets the user's input and builds a lightweight citizen context. Second, a decision matrix is used to assess the request. Low-risk requests, such as finding the correct office or explaining a public webpage, may be handled automatically. Medium-risk requests may require stronger source grounding and transparency. High-risk requests, such as residence status or legally sensitive benefit eligibility, require human review or strict limitation of the AI's role. Prohibited cases, such as unauthorized profiling, discriminatory recommendations, or final legal decisions without human oversight, must be blocked.

Once classified, requests are routed to appropriate existing AI tools for tasks such as service search, chatbots, document summarization, translation, and information retrieval. If no suitable tool exists, CivicTwin provides official source links, refers the case to human offices, or flags the gap for future development, enhancing transparency and sustainability.

CivicTwin also maintains database layers for user-consented profiles, official service data, tool capabilities, interaction logs, feedback, and audit records. IT Munich/DAICE oversees governance, maintains the decision matrix, reviews high-risk cases, monitors performance, and ensures that AI-supported services are legal, fair, and accountable.

CivicTwin addresses two core challenges: it transforms fragmented bureaucracy into a personalized experience for citizens and coordinates disparate AI tools into a unified, governable system for the city. Its novelty lies in combining a civic digital twin, an AI orchestration layer, and a governance decision matrix. To explore our prototype interactively, please scan the QR code.

3.5 Prototype Structure and Components

Layer	Role in CivicTwin	Example Outputs
Input	Citizen profile, question, language, location, consent mode, and current administrative goal.	Personalized context for service guidance.
Decision Matrix	Cross-system risk classification based on data sensitivity, task sensitivity, legal impact, fairness, transparency, accountability, sustainability, efficiency, and citizen impact.	Low/medium/high/prohibited handling path.
AI Toolbox	Routes requests to suitable existing AI tools and official data sources.	Search, explanation, summarization, translation, and chatbot support.
Human Oversight	IT Munich / DAICE reviews high-risk cases, audits performance, and maintains governance rules.	Escalation, review, monitoring, accountability.
Output	Creates citizen-facing guidance grounded in official sources.	Checklist, map, document list, official links, next action.

3.6 How CivicTwin Differs from Existing Solutions

Aspect	Current State	CivicTwin
Service Access	Fragmented across portals	Unified, personalized dashboard
Guidance	Citizens must search and guess	Proactive, step-by-step action plans
Personalisation	Generic, one-size-fits-all	Tailored to the citizens' profile and needs
Data Integration	Siloed across departments	Unified via digital twin architecture
Human Oversight	Ad-hoc	Systematic human-in-the-loop for legal effects
Sustainability	Not measured	Explicit CO_2e tracking and reduction

4.0 Discussion and Outlook

4.1. Target Groups

Newcomers are the primary target users of CivicTwin. This includes international students, skilled workers, expatriates, refugees, and individuals relocating to Munich. These users often face challenges such as unfamiliar administrative procedures, language barriers, fragmented government services, and limited knowledge of local regulations. CivicTwin simplifies the onboarding experience by providing a personalized, AI-powered digital companion that guides users through essential tasks, including city registration, residence permits, health insurance, banking, transportation, and other public services. By offering multilingual support and step-by-step guidance, CivicTwin helps newcomers settle into Munich more quickly and confidently. Our long-term goal is to scale vertically; as such, we target secondary groups of long-term residents who regularly access public services, renew permits, receive city updates, or participate in community activities. Moreover, Families require support for childcare, education, healthcare, housing, and other municipal services through one integrated platform.

4.2 Impact on Citizens and Government (Munich)

CivicTwin generates impact primarily through orchestration: a coordination layer interprets citizen needs, consults official sources, routes requests to existing municipal AI tools, and delivers a unified action plan. This approach aligns with the citizen-centric digital twin model, which emphasizes platforms that actively engage citizens in municipal processes (Luca et al., 2024).

For citizens, CivicTwin reduces administrative complexity through a single point of contact, facilitating access for newcomers, long-term residents, and families. For the government, automating routine processing reallocates caseworker capacity to higher-risk domains, consistent with a risk-based governance model aligned with the EU Artificial Intelligence Act (European Commission, 2024).

4.3 Sustainability Improvements

CivicTwin aligns with Munich's sustainability goals, notably SDGs 10, 11, 13, and 16 (United Nations, 2015). Sustainability is advanced by consolidating services to reduce travel and paper use, fostering institutional trust through transparent governance, and reusing existing tools to minimize computational and maintenance overhead (Luo et al., 2025).

4.4 Limitations

The project faces several limitations. The maturity of municipal AI tools varies, which constrains the reliability of orchestration. Accurate risk classification remains challenging due to regulatory ambiguity. Equity concerns persist for less digitally literate or non-native users. Integrating personal and municipal data also raises unresolved GDPR-related issues regarding consent and data minimization. These limitations suggest the following next steps: conducting a formal Data Protection Impact Assessment, beginning rollout with low-risk use cases, implementing oversight and audit procedures in accordance with AI Act Article 9, and co-designing accessibility with diverse user groups.

5.0 Conclusion

CivicTwin Munich illustrates how an orchestration-based civic digital twin can reduce administrative burden for citizens and free up administrative capacity for government, while its sustainability case rests on consolidating, trusting, and reusing existing infrastructure rather than new technology alone. Its principal risks lie in uneven component maturity, classification ambiguity, and unresolved equity and data-protection questions, which a phased, audited, and co-designed rollout could substantially mitigate.

6.0 Author Contribution

We began this study with a team assessment and assigned tasks to address the problem collaboratively. In the first week, we distributed responsibilities based on individual strengths and interests. Gulsah Deniz led literature reviews and data storytelling. Richmond Baidoo managed project structure and prototype implementation. Yuanqi Cheng developed the prototype concept and architecture. Ritthaisong Korait handled prototype design, layering, and final outputs. Mohammadisharmeh Samaneh worked on the decision matrix and data analysis. Rexford Asiedu evaluated overall impacts, focusing on sustainability measures supported by data reviews and questionnaires. All these roles inherently led to this report in writing.

References

Bitkom. (2024). *Smart City Index 2024: Munich remains in first place and extends its lead*. Smart Country Convention. https://www.smartcountry.berlin/en/newsblog/smart-city-index-2024-munich-remains-in-first-place-and-extends-its-lead.html

European Commission. (2024). *Regulation (EU) 2024/1689 of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)*. Official Journal of the European Union.

European Commission. (n.d.). *AI Act*. Shaping Europe's Digital Future. Retrieved June 25, 2026, from https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

European Data Protection Board. (2018). *Guidelines on automated individual decision-making and profiling for the purposes of Regulation 2016/679*. https://www.edpb.europa.eu/our-work-tools/our-documents/guidelines/automated-decision-making-and-profiling_en25

European Parliament & Council of the European Union. (2016). *Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data (General Data Protection Regulation)*. Official Journal of the European Union.

European Parliament & Council of the European Union. (2024). *Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence*. Official Journal of the European Union. https://eur-lex.europa.eu/eli/reg/2024/1689/oj

Fuller, A., Fan, Z., Day, C., & Barlow, C. (2020). Digital twin: Enabling technologies, challenges and open research. *IEEE Access, 8*, 108952–108971. https://doi.org/10.1109/ACCESS.2020.2998358

Key Issue 3 – Risk-Based Approach. (n.d.). *EU AI Act*. Retrieved June 22, 2026, from https://www.euaiact.com/key-issue/3

Landeshauptstadt München. (n.d.-a). *Bürgerservice-Suche*. Retrieved June 25, 2026, from https://stadt.muenchen.de/service/

Landeshauptstadt München. (n.d.-b). *KI-Systeme*. Retrieved June 25, 2026, from https://ki.muenchen.de/ki-systeme/

Luca, M., Lepri, B., Gallotti, R., Paolazzi, S., Bigi, M., & Pistore, M. (2024). *Towards civic digital twins: Co-design the citizen-centric future of Bologna* (arXiv:2412.06328). arXiv. https://arxiv.org/abs/2412.06328

Luo, J., Liu, P., Kong, X., Shen, J., Wu, Q., & Xu, D. (2025). Urban digital twins for citizen-centric planning: A systematic review of built environment perception and public participation. *International Journal of Applied Earth Observation and Geoinformation, 143*, Article 104746. https://doi.org/10.1016/j.jag.2025.104746

OECD. (n.d.). *AI principles*. Retrieved June 25, 2026, from https://www.oecd.org/en/topics/ai-principles.html

Rohde, F., Wagner, J., Meyer, A., Reinhard, P., Voss, M., Petschow, U., & Mollen, A. (2023). *Broadening the perspective for sustainable AI: Comprehensive sustainability criteria and indicators for AI systems*. arXiv. https://arxiv.org/abs/2306.13686

United Nations. (2015). *Transforming our world: The 2030 agenda for sustainable development (A/RES/70/1)*.

Zimmer, T. (2023). Faceting the post-disaster built heritage reconstruction process within the digital twin framework for Notre-Dame de Paris. *Scientific Reports, 13*, 5981. https://doi.org/10.1038/s41598-023-32504-9